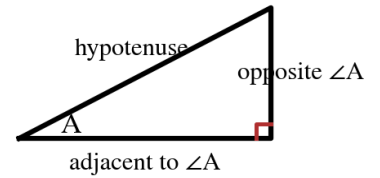


## Speak Like A Mathematician:

trigonometric ratio:

sine · cosine · tangent:

opposite · adjacent (relative to an angle):



### **SOH - CAH - TOA**

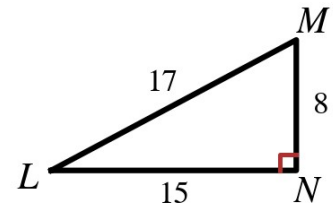
$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$  ·  $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$  ·  $\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$

**To find a missing side: 1) label the sides opp / adj / hyp from the given angle, 2) pick the ratio that uses the side you \_\_\_\_\_ and the side you \_\_\_\_\_, 3) set up the equation and solve.**

### **Example 1 - Write the Six Ratios**

In  $\triangle LMN$  at the right, the right angle is at N,  $LN = 15$ ,  $NM = 8$ , and  $LM = 17$ . Write each ratio as a fraction and a four-place decimal.

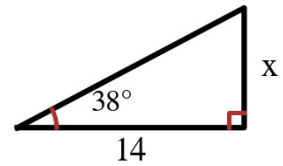
- a)  $\sin L$ ,  $\cos L$ ,  $\tan L$
- b)  $\sin M$ ,  $\cos M$ ,  $\tan M$
- c) What do you notice about  $\sin L$  and  $\cos M$ ?



### Example 2 - Find a Missing Side (worked)

In the triangle at the right, the  $38^\circ$  angle has adjacent side 14 and opposite side  $x$ .

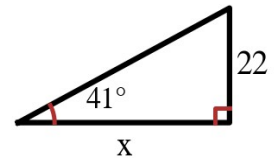
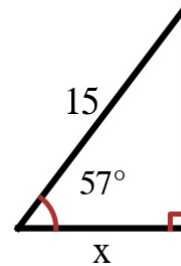
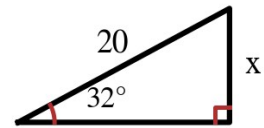
- Which ratio applies, and why?
- Set up and solve. Round to the nearest tenth.



### Example 3 - Find the Missing Side

Round each answer to the nearest tenth. Figures at the right.

- $32^\circ$  angle, hypotenuse 20, opposite  $x$ .
- $57^\circ$  angle, hypotenuse 15, adjacent  $x$ .
- $41^\circ$  angle, opposite 22, adjacent  $x$ .



### Example 4 - A Preview of Inverse Trig

You stand 275 ft from a 600-ft skyscraper (figure at the right).

- Write the tangent of the angle of elevation  $x^\circ$ .
- Find  $x$  with the  $\tan^{-1}$  key. Round to the nearest tenth.

